

C Programming Practice Exam

Topics: Strings, Pointers, Structs, Linked Lists

Total Time: 60 minutes

Total Points: 40

Question 1 — String Manipulation (10 pts)

Write a function that takes a string and reverses it **in place** using only pointers (no array indexing). You may not use `strlen()`.

```
void reverse_string(char *s);
```

Example:

Input: "hello"

Output: "olleh"

Write your implementation below:

Question 2 — Structs and Dynamic Memory (10 pts)

You are given:

```
struct Product {  
    char *name;  
    double price;  
    struct Product *next;  
};
```

Write a function that: 1. Allocates a new `Product`. 2. Copies a given name string into dynamically allocated memory. 3. Sets its price. 4. Returns a pointer to the new product.

```
struct Product* create_product(const char *name, double price);
```

Question 3 — Linked List Manipulation (10 pts)

You are given a singly linked list of `Products` as above. Write a function that removes the **first node with price less than 5.0** and frees its memory. Return the (possibly new) head of the list.

```
struct Product* remove_cheap(struct Product *head);
```

Question 4 — Short Answer (10 pts)

1. (3 pts) What is a segmentation fault, and what typically causes it in C?
2. (3 pts) What is the output of the following code?

```
int x = 5;
int *p = &x;
printf("%d\n", *p + 1);
```

3. (4 pts) Consider:

```
struct Node { int data; struct Node *next; };
struct Node a = {1, NULL};
struct Node b = {2, &a};
struct Node *head = &b;
printf("%d\n", head->next->data);
```

What value is printed and why?